

A PROOF OF THE BHAGAT–KULKARNI–LARSSON–MURALI SMALLEST-DISCREPANCY-HEAP FORMULA

ABSTRACT. In the two-player self-interest cumulative subtraction game on the subtraction set $S = \{s_2, s_1\}$, $0 < s_2 < s_1$, the mover takes some $s \in S$ with $s \leq h$ from a heap of size h and adds s to their own score. Indifferent positions must be resolved by a convention: under FvF the mover breaks a tie in the opponent's favour, under AvA against it. Bhagat, Kulkarni, Larsson and Murali conjectured that in the non-dominant regime $s_1 < 2s_2$ with $s_1/s_2 \neq (k+1)/k$ for every $k \in \mathbb{N}$ some heap distinguishes the two conventions, and that the smallest such heap is $h = (n+2)s_2 + ns_1$, where $n = \lfloor s_2/(s_1 - s_2) \rfloor$. We prove both assertions. We exhibit, in closed form, the pure subgame perfect equilibrium outcome at *every* heap below that value—the same in both conventions—and evaluate both conventions at the value itself, obtaining $o_{\text{FvF}} = (s_2 + ns_1, (n+1)s_2)$ and $o_{\text{AvA}} = (s_2 + ns_1, ns_1)$. Existence and minimality are then immediate, the mover's coordinate agrees at the distinguishing heap, and the whole discrepancy is $-(s_2 \bmod (s_1 - s_2))$ in the non-mover's coordinate. We claim no priority: a manuscript of Kulkarni and Larsson listed as in preparation in that paper is on exactly this subject and could not be obtained.

1. THE GAME AND THE CONJECTURE

Fix a finite $S \subset \mathbb{Z}_{>0}$. In the *self-interest cumulative subtraction game* on S two players alternate; from a heap of h tokens the mover removes some $s \in S$ with $s \leq h$ and adds s to their *own* tally; play ends when $h < \min S$. The game is not zero-sum, and the relevant solution concept is a pure subgame perfect equilibrium (PSPE). Write $o(h) = (o^1(h), o^2(h))$ for the PSPE payoff pair, o^1 being the mover's total. Then $o(h) = (0, 0)$ for $h < \min S$ and

$$(1) \quad o^1(h) = \max_{s \in S, s \leq h} (s + o^2(h - s)), \quad o^2(h) = o^1(h - s^*),$$

where s^* is the maximiser. When two moves are *indifferent*, i.e. they attain the same value of $s + o^2(h - s)$, (1) does not determine $o^2(h)$ and a convention is needed. Following Bhagat–Kulkarni–Larsson–Murali [1], in the *friendly* convention FvF the mover selects, among indifferent moves, one *maximising* $o^1(h - s)$, and in the *antagonistic* convention AvA one *minimising* it. Both

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recursions are well defined and single-valued; we write o_{FvF} and o_{AvA} , and set

$$\delta^1(h) = o_{\text{AvA}}^1(h) - o_{\text{FvF}}^1(h), \quad \delta^2(h) = o_{\text{AvA}}^2(h) - o_{\text{FvF}}^2(h).$$

Call h a *discrepancy heap* if $o_{\text{FvF}}(h) \neq o_{\text{AvA}}(h)$.

From here on $S = \{s_2, s_1\}$ with $0 < s_2 < s_1$, and it is convenient to put

$$(2) \quad a := s_2, \quad b := s_1, \quad d := b - a > 0.$$

The object of this note is the following conjecture of [1], which we quote verbatim, translating only its internal labels into words.

Conjecture 1 ([1]). Consider $S = \{s_2, s_1\}$ in the non-dominant regime where $s_1 < 2s_2$, and suppose, for all $k \in \mathbb{N}$, $s_1/s_2 \neq (k+1)/k$. Then there exists a heap h that distinguishes the outcomes of the FvF and AvA conventions. The smallest such heap is defined by

$$(3) \quad h = (n+2)s_2 + ns_1.$$

The index n is fixed in [1] by the interval condition $\frac{n+1}{n} > \frac{s_1}{s_2} > \frac{n+2}{n+1}$, and a remark there gives the equivalent closed form $n = \lfloor s_2/(s_1 - s_2) \rfloor$. The two agree, and their agreement already encodes the hypotheses:

Lemma 1. *With the notation (2), the hypotheses of Conjecture 1 are exactly $0 < d < a$ together with $d \nmid a$; and under them $n = \lfloor a/d \rfloor \geq 1$ is the unique integer with $n < a/d < n+1$.*

Proof. $s_1/s_2 = 1 + d/a$, so $\frac{n+1}{n} > 1 + \frac{d}{a}$ reads $a/d > n$ and $1 + \frac{d}{a} > \frac{n+2}{n+1}$ reads $a/d < n+1$. Non-dominance $s_1 < 2s_2$ is $d < a$, i.e. $a/d > 1$, i.e. $n \geq 1$. The excluded ratios are $s_1/s_2 = (k+1)/k$, i.e. $1 + d/a = 1 + 1/k$, i.e. $a = kd$; so excluding them is exactly $d \nmid a$, which is what makes both inequalities of the interval condition strict. $\square$

Throughout, therefore,

$$(4) \quad 0 < d < a, \quad d \nmid a, \quad n := \left\lfloor \frac{a}{d} \right\rfloor \geq 1, \quad \varepsilon := a - nd \in (0, d), \quad \alpha_t := a - td,$$

and the heap named by (3) is

$$(5) \quad H := (n+2)a + nb = (2n+2)a + nd.$$

The neighbouring regime is already settled in [1]: a proposition proved there forbids any discrepancy once $s_1 \geq 2s_2$. For Conjecture 1 [1] offer computational evidence, a figure reaching $n = 48$ and further simulations.

2. THE RESULT

Theorem 2. *Assume (4) and let H be as in (5). Call a pair of integers (m, i) admissible if $m \geq 0$ and $0 \leq i \leq \lfloor m/2 \rfloor$, and put*

$$v(m, i) := ia + (m - i)b = mb - id.$$

Let $W := \{v(m, i) : (m, i) \text{ admissible}\}$, let $V(h) := \max\{w \in W : w \leq h\}$, let (m, i) be the admissible coordinates of $V(h)$ (unique for $h \leq H$, by Lemma 4) and $t := \lfloor m/2 \rfloor$. Then for every $h < H$, in both conventions,

$$(6) \quad (o^1(h), o^2(h)) = \begin{cases} (tb, tb - id), & m = 2t, \\ ((t+1)b - id, tb), & m = 2t + 1, \end{cases}$$

while at $h = H$

$$(7) \quad o_{\text{FvF}}(H) = (a + nb, (n+1)a) \neq (a + nb, nb) = o_{\text{AvA}}(H).$$

Corollary 3. *Conjecture 1 is true. A discrepancy heap exists, the smallest one is exactly $H = (n+2)s_2 + ns_1$, and there $\delta^1(H) = 0$ while*

$$\delta^2(H) = -\varepsilon = -(s_2 \bmod (s_1 - s_2)) < 0.$$

Proof. By (6) the outcome at every $h < H$ is the same in both conventions, so no $h < H$ is a discrepancy heap; by (7), H is one. The two first coordinates in (7) are equal, and $nb - (n+1)a = n(a+d) - (n+1)a = nd - a = -\varepsilon$. $\square$

Equation (6) is a complete description of the equilibrium payoff and aggregate move composition below H : PSPE play lasts m moves and consumes i copies of s_2 and $m-i$ copies of s_1 , leaving the residue $h - V(h)$ on the heap. It does not single out an equilibrium strategy or a unique move sequence.

3. THE LATTICE W

Write $L_m := \{v(m, i) : 0 \leq i \leq \lceil m/2 \rceil\}$, an arithmetic progression of step d with

$$\max L_m = mb, \quad \min L_{2t} = t(a+b), \quad \min L_{2t+1} = (t+1)a + tb.$$

Lemma 4. *Assume (4). Then*

- (F1) $\min L_{2t+1} - \max L_{2t} = \alpha_t$ and $\min L_{2t+2} - \max L_{2t+1} = \alpha_t$ for every $t \geq 0$;
- (F2) $\alpha_t \geq \alpha_n = \varepsilon > 0$ for $t \leq n$, hence $L_0, L_1, \dots, L_{2n+2}$ are pairwise disjoint and increasing, and every $w \in W$ with $w \leq H$ has a unique admissible coordinate pair;
- (F3) $\min L_{2n+2} = H + d$ and $\max L_{2n+1} = H + (d - \varepsilon) > H$; consequently $W \cap [0, H] \subseteq L_0 \cup \dots \cup L_{2n+1}$, $H \notin W$, and $\max(W \cap [0, H]) = v(2n+1, 1) = 2nb + a = H - \varepsilon$;
- (F4) for admissible (m, i) with $m \leq 2n+1$ — which by (F3) covers every point of $W \cap [0, H]$ — the gap in W immediately above $v(m, i)$ is d if $i \geq 1$ and $\alpha_{\lfloor m/2 \rfloor}$ if $i = 0$.

Proof. (F1) $\min L_{2t+1} - \max L_{2t} = (t+1)a + tb - 2tb = (t+1)a - tb = a - td = \alpha_t$, using $b = a + d$; and $\min L_{2t+2} - \max L_{2t+1} = (t+1)(a+b) - (2t+1)b = (t+1)a - tb = \alpha_t$.

(F2) $\alpha_t = a - td$ is decreasing in t and $\alpha_n = a - nd = \varepsilon > 0$ by (4); so all the gaps in (F1) with $t \leq n$ are strictly positive, each L_m has step $d > 0$,

and $L_0 < L_1 < \dots < L_{2n+2}$ as sets of integers. Uniqueness of coordinates on $[0, H]$ follows, because by (F3) no $v(m, i)$ with $m \geq 2n+2$ lies in $[0, H]$. (Positivity of ε is exactly where $d \nmid a$ enters.)

(F3) $\min L_{2n+2} = (n+1)(a+b) = (2n+2)a + (n+1)d = H + d$, and $\max L_{2n+1} = (2n+1)b = (2n+1)a + (2n+1)d = H + ((n+1)d - a) = H + (d - \varepsilon)$, which exceeds H since $\varepsilon < d$. As $\min L_m$ is increasing in m , every $v(m, i)$ with $m \geq 2n+2$ is at least $H + d$, which gives the inclusion. Inside L_{2n+1} , $v(2n+1, i) \leq H$ reads $(1-i)d \leq \varepsilon$, true for $i \geq 1$ and false for $i = 0$; so the largest element of W below H is $v(2n+1, 1) = (2n+1)b - d = 2nb + a$, at distance $H - (2nb + a) = (2n+2)a + nd - (2n+1)a - 2nd = \varepsilon$ from H . Since $0 < \varepsilon < d$, that distance is not a multiple of d , so $H \notin L_{2n+1}$; and $H < \min L_{2n+2}$ while $H > \max L_{2n} = 2nb$, so $H \notin W$.

(F4) Let $m \leq 2n+1$. Since $\min L_{m'}$ is increasing in m' , every element of W lying outside $L_0 \cup \dots \cup L_{2n+2}$ is at least $\min L_{2n+3} > \min L_{2n+2}$; so on $[0, \min L_{2n+2}]$ the elements of W are exactly those of $L_0 \cup \dots \cup L_{2n+2}$, ordered as in (F2). Both $v(m, i)$ and $\min L_{m+1}$ lie in that range, since $\max L_{2n+1} < \min L_{2n+2}$ by (F3). If $i \geq 1$ then the next element of L_m , and hence of W , is $v(m, i-1) = v(m, i) + d$. If $i = 0$ then $v(m, 0) = \max L_m$ and the next element of W is $\min L_{m+1}$, at distance $\alpha_{\lfloor m/2 \rfloor}$ by (F1). $\square$

4. PROOF OF THEOREM 2

We induct on h . For $h < a$ the only admissible value $\leq h$ is $v(0, 0) = 0$, so $m = i = t = 0$ and (6) reads $o(h) = (0, 0)$, which is the base of (1).

Let $a \leq h \leq H$ and let (m, i) be the coordinates of $V(h)$, $t = \lfloor m/2 \rfloor$, $r := h - V(h)$. Since $v(1, 1) = a \leq h$ we have $m \geq 1$, and by (F3) $m \leq 2n+1$, hence

$$(8) \quad t \leq n, \quad \text{so} \quad \alpha_t \geq \varepsilon > 0.$$

Put $c_a := a + o^2(h - a)$ and, when $h \geq b$, $c_b := b + o^2(h - b)$; by (1) the mover takes the larger, and on a tie FvF takes the s maximising $o^1(h - s)$ while AvA takes the s minimising it. By admissibility $0 \leq i \leq t$ when $m = 2t$ and $0 \leq i \leq t+1$ when $m = 2t+1$, so exactly one of the following five cases holds; the split is therefore exhaustive and disjoint. In each case we use the inductive hypothesis (6) at $h - a$ and $h - b$, whose coordinates we first locate using (F4).

Case 1: $m = 2t$, $1 \leq i \leq t$. Here $r < d$ by (F4). Since $b - a = d$ and $v(m, i) - b = v(2t-1, i)$, $v(m, i) - a = v(2t-1, i-1)$, and since the gaps above $v(2t-1, i)$ and $v(2t-1, i-1)$ are d (or $\alpha_{t-1} = \alpha_t + d > d$ when $i = 1$) we get $V(h - b) = v(2t-1, i)$ and $V(h - a) = v(2t-1, i-1)$. Both have $m' = 2t-1$ odd with $t' = t-1$, so (6) gives $o^2(h - b) = o^2(h - a) = (t-1)b$ and

$$c_b - c_a = b - a = d > 0.$$

Thus $s^* = s_1$ uniquely, $o^1(h) = c_b = tb$ and $o^2(h) = o^1(h - b) = tb - id$: this is (6) for $m = 2t$, in both conventions.

Case 2: $m = 2t$, $i = 0$, $t \geq 1$. Here $r < \alpha_t$ by (F4). Now $h - b = v(2t - 1, 0) + r$ and $h - a = v(2t - 1, 0) + d + r$, and $d + r < d + \alpha_t = \alpha_{t-1}$, so both residues sit on $v(2t - 1, 0)$ and $o^2(h - a) = o^2(h - b) = (t - 1)b$. Again $c_b - c_a = d > 0$, so $s^* = s_1$ uniquely, $o^1(h) = tb$ and $o^2(h) = o^1(h - b) = tb$, which is (6) with $i = 0$.

Case 3: $m = 2t + 1$, $1 \leq i \leq t$ (so $t \geq 1$). Here $r < d$. As $v(m, i) - b = v(2t, i)$ with $i \geq 1$, we get $V(h - b) = v(2t, i)$, $o^2(h - b) = tb - id$, $o^1(h - b) = tb$ and

$$c_b = (t + 1)b - id.$$

Also $v(m, i) - a = v(2t, i - 1)$ exactly. Two sub-cases, according to (F4) applied at $v(2t, i - 1)$, whose gap above is d if $i \geq 2$ and α_t if $i = 1$.

Case 3A: $i \geq 2$, or $i = 1$ and $r < \alpha_t$. Then $V(h - a) = v(2t, i - 1)$, so $o^2(h - a) = tb - (i - 1)d$ and $c_a = a + tb - (i - 1)d = (t + 1)b - id = c_b$: the two moves are indifferent. But the tie-breaking comparands agree, $o^1(h - a) = tb = o^1(h - b)$ by (6), so both conventions return the same pair $((t + 1)b - id, tb)$ — the tie is not load-bearing.

Case 3B: $i = 1$ and $r \geq \alpha_t$. Since $r < d$ this forces $\alpha_t < d$, i.e. $a < (t + 1)d$, i.e. $t \geq n$; with (8), $t = n$. Hence $\alpha_t = \varepsilon$, $V(h) = v(2n + 1, 1) = H - \varepsilon$ by (F3), and $r \geq \varepsilon$ forces $h \geq H$, so $h = H$. *Case 3B therefore occurs at $h = H$ and nowhere below it;* this is the minimality assertion. At $h = H$ we have $r = \varepsilon$ and

$$H - a = (2n + 1)a + nd = v(2n + 1, n + 1), \quad H - b = v(2n, 1) + \varepsilon,$$

so by (6)

$$o^1(H - a) = (n + 1)b - (n + 1)d = (n + 1)a, \quad o^2(H - a) = nb,$$

$$o^1(H - b) = nb, \quad o^2(H - b) = nb - d.$$

Hence $c_a = a + nb$ and $c_b = b + nb - d = (n + 1)a + nd = a + nb = c_a$: again a tie. This time the comparands *split*,

$$o^1(H - a) - o^1(H - b) = (n + 1)a - nb = a - nd = \varepsilon > 0.$$

So FvF, which maximises the opponent's payoff, plays s_2 and the non-mover receives $o^1(H - a) = (n + 1)a$; AvA plays s_1 and the non-mover receives $o^1(H - b) = nb$. Both give $o^1(H) = a + nb$. This is (7).

Case 4: $m = 2t + 1$, $i = 0$. Here $r < \alpha_t$ by (F4), and $t \leq n - 1$: indeed $v(2n + 1, 0) > H$ by (F3). Now $h - b = v(2t, 0) + r$ with $r < \alpha_t$, so $V(h - b) = v(2t, 0)$, $o^2(h - b) = tb$ and $c_b = (t + 1)b$. For $h - a = v(2t, 0) + d + r$ there are two possibilities — if $d + r < \alpha_t$ then $V(h - a) = v(2t, 0)$, and otherwise $v(2t, 0) + \alpha_t \leq h - a < v(2t, 0) + \alpha_t + d$ so $V(h - a) = v(2t + 1, t + 1)$ — but (6) returns $o^2(h - a) = tb$ in *either* case, so $c_a = a + tb$ and

$$c_b - c_a = b - a = d > 0.$$

Thus $s^* = s_1$ uniquely, $o^1(h) = (t + 1)b$ and $o^2(h) = o^1(h - b) = tb$, which is (6).

Case 5: $m = 2t + 1$, $i = t + 1$. Then $V(h) = \min L_{2t+1} = (t + 1)a + tb$ and $r < d$. If $t = 0$ then $a \leq h < b$ and only s_2 is legal, giving $o(h) = (a, 0)$, which is (6). If $t \geq 1$, then $h - a = v(2t, t) + r$ with $t \geq 1$, so $V(h - a) = v(2t, t)$ and $o^2(h - a) = tb - td = ta$, whence $c_a = (t + 1)a$; and $h - b = v(2t - 1, 0) + \alpha_t + r$ with $\alpha_t + r < \alpha_t + d = \alpha_{t-1}$, so $V(h - b) = v(2t - 1, 0)$ and $o^2(h - b) = (t - 1)b$, whence $c_b = tb$. Therefore

$$c_a - c_b = (t + 1)a - tb = \alpha_t > 0$$

by (8): the *sacrifice* $s^* = s_2$ is uniquely optimal. Then $o^1(h) = (t + 1)a = (t + 1)b - (t + 1)d$ and $o^2(h) = o^1(h - a) = tb$, which is (6) with $i = t + 1$.

Every case with $h < H$ — that is, Cases 1, 2, 3A, 4, 5 — produces a convention-independent pair satisfying (6), and $h = H$ falls in Case 3B and produces (7). This completes the induction and the proof. $\square$

Remark (where the hypotheses are used). The two hypotheses enter together, through $\varepsilon > 0$: non-dominance $d < a$ gives $n \geq 1$, and $d \nmid a$ then gives $\varepsilon = a - nd > 0$, hence $\alpha_t \geq \varepsilon > 0$ for $t \leq n$ in (8), hence the strict sacrifice of Case 5 and the strict comparand gap ε at H . At $d \geq a$ the ordering of Lemma 4 collapses and the proposition of [1] recalled in §1 takes over; at $\varepsilon = 0$ Case 3B degenerates into Case 3A and that tie is not load-bearing.

5. AN EXAMPLE

The hypothesis set $\{(a, d) : 1 \leq d < a, d \nmid a\}$ is non-empty from $a = 3$, $d = 2$, i.e. $S = \{3, 5\}$, which is the running example of [1]. Here $n = 1$, $\varepsilon = 1$ and $H = 3 \cdot 3 + 1 \cdot 5 = 14$. The admissible values ≤ 13 are 0, 3, 5, 8, 10, 11, 13 with coordinates (0, 0), (1, 1), (1, 0), (2, 1), (2, 0), (3, 2), (3, 1), and (6) gives the block table

h	[0, 3]	[3, 5]	[5, 8]	[8, 10]	[10, 11]	[11, 13]	[13, 14]
(o^1, o^2)	(0, 0)	(3, 0)	(5, 0)	(5, 3)	(5, 5)	(6, 5)	(8, 5)

while (7) gives $o_{\text{FvF}}(14) = (8, 6)$ and $o_{\text{AvA}}(14) = (8, 5)$. The table for $S = \{3, 5\}$ in [1] prints the columns $h = 0, \dots, 15$ with $\delta^1 \equiv 0$ and the single non-zero entry $\delta^2(14) = -1$, and records the PSPE move sequences 3-3-5-3 under FvF and 5-5-3 under AvA, which is Case 3B's s_2 against s_1 .

6. SCOPE AND VERIFICATION

What is proved is Conjecture 1 in full, uniformly in $n \geq 1$. Not settled: the companion statement of [1], that at the excluded ratios $s_1/s_2 = (k+1)/k$ there is no discrepancy at any heap—nothing here bears on it, and at $d \mid a$ one has $\alpha_n = 0$, Lemma 4(F2) fails at $t = n$ and the coordinates of Theorem 2 are not even well defined, so in particular this proof does not show the absence of a discrepancy at or below H there; heaps $h > H$, where the outcome genuinely is convention-dependent and no claim is made; subtraction sets with $|S| \geq 3$; and formalisation, this being a hand proof, not checked in a proof assistant. Reference [12] of [1] is [4], by two of its four authors and on this subject, listed there as *in preparation*, with no

arXiv identifier and no DOI; we were unable to obtain it, so we do not know what it contains. We claim no novelty or priority against it, nor against [2], whose solution of the two-action *zero-sum* game already contains part of the structure of play below H (a zero-sum game has a single-valued outcome and so no conventions to disagree, so [2] states neither half of Conjecture 1), nor against [3], where the instance $S = \{3, 5\}$ at $h = 14$ is on record.

A program accompanying this note, `verify.py`, with a transcript of one run in `verify.output.txt`, brute-forces (1) in both conventions in exact integer arithmetic over the 4568 pairs with $s_2 \leq 100$ satisfying (4), and checks the invariant (6) and its convention-independence at every $h < H$, the values (7) and $\delta^2(H) = -\varepsilon$, that the least discrepancy heap is exactly H , each part of Lemma 4, and the exhaustiveness, disjointness and per-case arithmetic of the five-case split of §4; all 32 of its checks pass. Since $d \nmid a$ forces $d \geq 2$, that box reaches only $n \leq 49$, so for the formula itself it adds little to the computational evidence already in [1]; what it adds is the check of the structural claims. It audits the induction on a finite range and does not certify Theorem 2, which is proved above by hand.

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